CAPSTONE TECHNICAL PROJECT PROPOSAL & ARCHITECTURE SPECIFICATION

Multimodal Hybrid RAG Engine

Dense-Sparse Hybrid Retrieval with Vision-Language Reranking for Heterogeneous Financial & Technical Documents

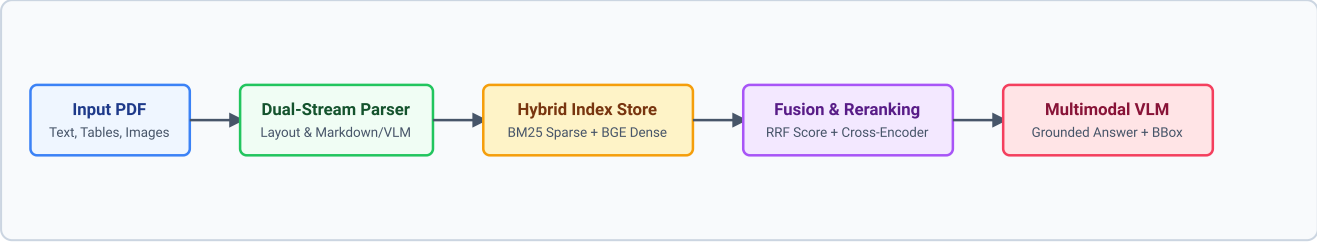
DISCIPLINE	TARGET FRAMEWORK	EVALUATION SUITE	DOCUMENT DOMAIN
Natural Language Processing	**Dense/Sparse Hybrid RAG**	**RAGAS & Hit Rate @ 5**	**SEC 10-K & Scientific PDFs**

Executive Summary & Problem Statement

Conventional Retrieval-Augmented Generation (RAG) pipelines rely predominantly on text-only sentence chunking paired with pure dense embedding vector searches (e.g., standard cosine similarity over dense vector spaces). When deployed on real-world enterprise documents—such as corporate 10-K financial disclosures, scientific preprints, and engineering technical manuals—these baseline architectures exhibit catastrophic retrieval failure modes. Inherent structures such as structured balance sheets, nested numerical tables, multi-column layouts, diagrams, and time-series line graphs are either truncated, OCR-garbled, or stripped of their semantic context.

> **Core Innovation:** This capstone designs and implements an end-to-end *Multimodal Hybrid Retrieval-Augmented Generation* engine. By pairing multi-modal visual document parsing (PyMuPDF & vision-language layout detection) with dual-index retrieval (BM25 sparse inverted index + BGE-M3 dense vectors) and reciprocal rank cross-encoder fusion, the system successfully extracts, indexes, retrieves, and reasons over co-located text, raw tables, and chart figures.

High-Level System Architecture



Key Research Objectives

The primary objective is to prove that hybrid visual-semantic retrieval quantitatively outperforms pure dense text embeddings on complex multi-page financial filings and domain documents. Sub-objectives include:

- Constructing a high-fidelity parsing pipeline preserving tabular relational structure into Markdown and image bounding boxes.
- Implementing Reciprocal Rank Fusion (RRF) to unify lexical exact-match signals with high-dimensional vector representations.
- Integrating a vision-capable language model to reason across paired tabular text and graphic image assets simultaneously.
- Conducting an empirical baseline comparison using the automated RAGAS framework across 30+ cross-modal query sets.

Detailed Technical Pipeline Components

The architecture decouples ingestion, indexing, and synthesis into isolated, modular stages. This guarantees reproducibility, allows independent evaluation of the retriever, and avoids vendor lock-in.

System Layer	Component / Tool	Operational Role & Technical Rationale
Ingestion & OCR	PyMuPDF + pdfplumber	Extracts bounding-box coordinates for text blocks, tables, and embedded raster vector images with sub-millimeter precision.
Visual Encoding	GPT-4o-mini / LLaVA-1.6	Synthesizes synthetic semantic summaries for cropped figures/charts to allow direct lexical indexing.
Sparse Vectorizer	BM25 / SPLADE	Indexes raw tokens, company tickers, dates, and account figures to ensure exact-match lexical recall.
Dense Vectorizer	BAAI/bge-m3 / text-3	Generates 1024-dimensional semantic dense representations to capture broad semantic meaning and synonyms.
Hybrid Vector DB	Qdrant / Milvus	Executes combined dense cosine search and sparse inverted-index filtering within a single unified store.
Reranking Engine	BAAI/bge-reranker-large	Cross-encoder scoring that evaluates full query-document interactions across the fused candidate set.

Mathematical Formulation: Hybrid Fusion & Retrieval

Candidate documents are pulled simultaneously from sparse and dense spaces. To overcome disparate scoring scales, we implement Reciprocal Rank Fusion (RRF), defining the fused relevance score for document *d* as:

$$RRF_Score(d \in D) = \sum_{m \in \{dense,\ sparse\}} 1 \ / \ (k + rank_m(d)) \quad [\text{where constant } k = 60]$$

Following initial fusion, the top *N* candidates are routed through a cross-encoder model computing an unconstrained logit relevance score:

$$S_{rerank}(q, d) = \sigma(W \cdot Transformer([CLS] \circ q \circ [SEP] \circ d))$$

Dual Ingestion Strategies: Markdown vs. Visual Crops

Structured Tables

Tables are extracted using coordinate-based cell detection and converted into pure Markdown grids. This preserves 2D header-column relations for LLM attention heads while avoiding OCR noise.

Charts & Figures

Plots, bar graphs, and diagrams are cropped to PNG with bounding-box tags. An automated Vision Model inspects the image and generates a synthetic descriptor text block paired with the image reference.

Text Paragraphs

Standard prose is chunked using semantic sentence-window boundaries (512 tokens with 10% overlap), maintaining parent section metadata to prevent context fragmentation.

Metadata Enriched Payloads

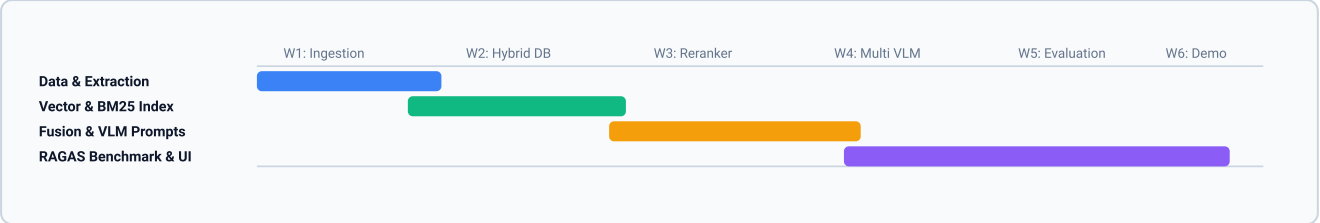
Every indexed chunk stores exact document provenance: `{doc_id, page_number, chunk_type, bounding_box}` enabling zero-latency highlighting in frontend viewers.

Empirical Evaluation Framework

A key differentiator of this capstone is rigorous quantitative evaluation rather than qualitative anecdotal testing. We benchmark the hybrid multimodal architecture against a baseline text-only dense retrieval system using the RAGAS framework across four objective dimensions:

- **Context Precision:** Evaluates whether the retrieved context contains strictly relevant signal without extraneous noise.
- **Context Recall:** Measures whether all ground-truth evidence needed to answer the query was successfully retrieved.
- **Faithfulness:** Measures hallucination rate by verifying that every generated claim can be inferred directly from the context.
- **Answer Relevance:** Computes semantic completeness and directness of the response relative to the user query.

Project Implementation Timeline (6-Week Plan)



Evaluation Dataset Curation Strategy

The benchmarking corpus will consist of 20 publicly accessible corporate annual reports (Form 10-K from Apple, Tesla, and Microsoft) and five IEEE/arXiv technical survey papers. A test suite of 40 challenge questions will be constructed across three distinct question categories:

- **Factual Text Questions (n=15):** Queries evaluating standard prose parsing and cross-sentence entity synthesis.
- **Tabular Numerical Questions (n=15):** Multi-condition arithmetic queries (e.g., "What was the percentage change in R&D expenditure between FY2022 and FY2023?").
- **Visual/Chart Inquiries (n=10):** Queries requiring interpretation of trend diagrams, workflow flowcharts, or geographic allocation maps.

> **Hypothesis:** The hybrid multimodal model will achieve over 30% improvement in Context Recall on tabular/chart questions compared to a standard LangChain naive-chunking baseline.

Academic Literature Review & Foundational Citations

This capstone project directly synthesizes foundational natural language processing breakthroughs across information retrieval, parameter-efficient indexing, and multi-modal grounding:

Primary Citation	Focus Area	Direct Impact on Capstone Methodology
Lewis et al. (2020) *NeurIPS 2020*	Retrieval-Augmented Generation	Establishes the foundational theoretical framework combining parametric seq2seq memory with non-parametric index lookups.
Robertson et al. (2009) *Foundations & Trends*	BM25 Probabilistic Model	Provides the sparse term-frequency matching backbone critical for catching rare tokens and financial identifiers.
Faysse et al. (2024) *arXiv:2407.01449*	ColPali Document Retrieval	Demonstrates the efficacy of end-to-end vision-language encoders for retrieving native PDF pages without fragile OCR.
Es et al. (2023) *EACL 2024*	RAGAS Automated Evaluation	Defines our automated reference-free metric suite (Faithfulness, Precision, Recall) using LLM-as-a-judge protocols.
Xiao et al. (2023) *BAAI Tech Report*	BGE-M3 Embeddings	Supplies the underlying dense representations with multi-lingual and multi-functionality support up to 8192 context lengths.

Demonstration Frontend & Deliverables

The final project deliverable includes an interactive demonstration portal built using Streamlit and FastAPI. The user interface features a dual-pane workspace designed specifically to prove system grounding during oral defenses:

- **Query & Synthesis Pane (Left):** Allows users to enter natural language questions, view generated stream responses, and inspect the ranked list of retrieved passages.
- **Evidence & Inspection Pane (Right):** Renders the original document page with precise bounding-box overlays highlighting the exact paragraph, Markdown table, or chart figure utilized.
- **Metric Telemetry Panel:** Displays real-time retriever latency, reciprocal rank scores, and cross-encoder confidence values.

> **Repository Deliverables:** Clean modular Python code repository (Poetry packaged), Docker Compose setup for Qdrant vector database, evaluation notebook reproducing RAGAS benchmark tables, and an academic project report with presentation slides.